

Set 7 Customer Demographics Analysis

Dataset: Customer Demographics

Customer ID Age Gender Income (in \$)

1 28 Female 50000

2 35 Male 60000

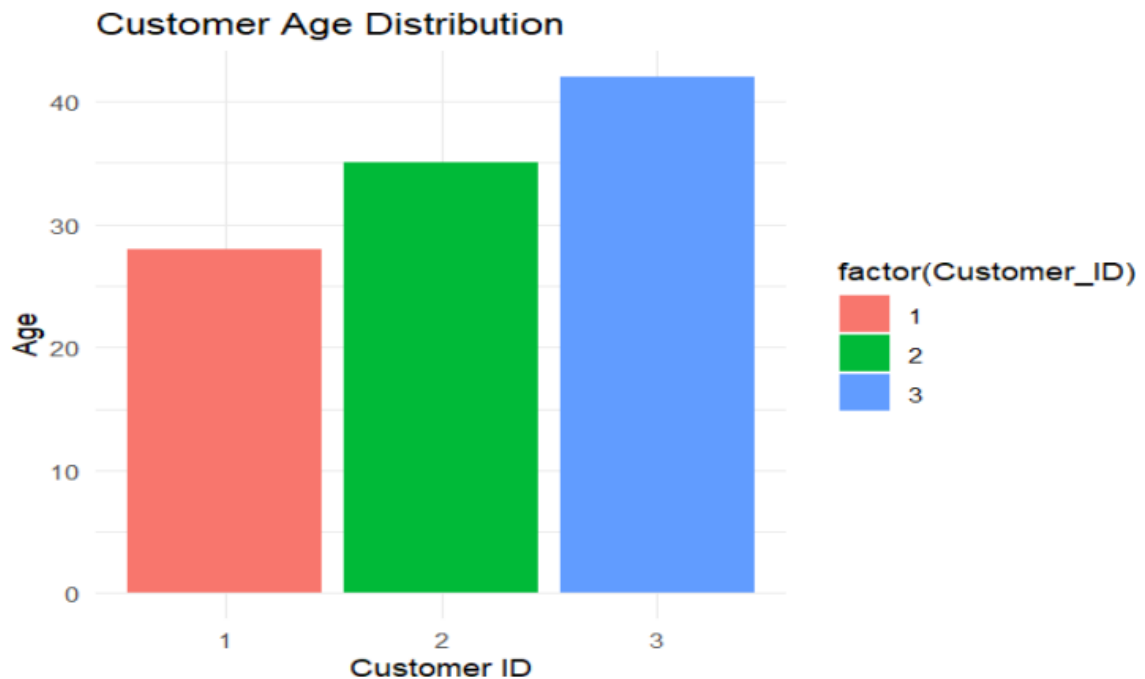
3 42 Female 75000

- 1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.**

```
# Load library
library(ggplot2)

# Create dataset
customers <- data.frame(
  Customer_ID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)

# Bar Chart
ggplot(customers,
  aes(x = factor(Customer_ID),
    y = Age,
    fill = factor(Customer_ID))) +
  geom_bar(stat = "identity") +
  labs(
    title = "Customer Age Distribution",
    x = "Customer ID",
    y = "Age"
  ) +
  theme_minimal()
```



2. Generate a pie chart to represent the distribution of customers by gender.

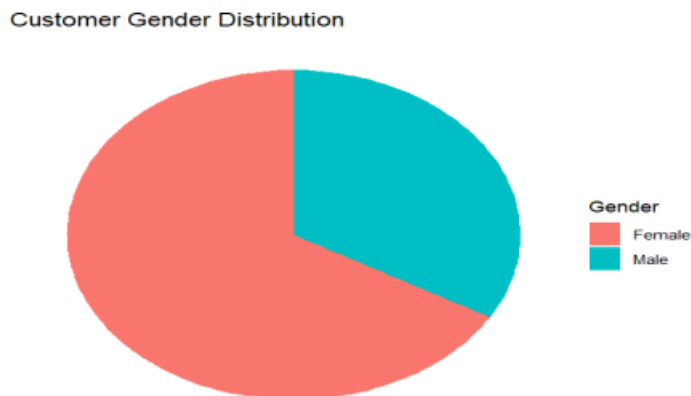
```
# Load libraries
library(ggplot2)

# Create dataset
customers <- data.frame(
  Customer_ID = c(1,2,3),
  Gender = c("Female","Male","Female")
)

# Count genders
gender_count <- as.data.frame(table(customers$Gender))
colnames(gender_count) <- c("Gender","Count")

# Pie Chart
ggplot(gender_count,
  aes(x = "",
    y = Count,
    fill = Gender)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y") +
```

```
labs(title = "Customer Gender Distribution") +
theme_void()
```



3. Build a table to show the demographic information for each customer. Label the table elements.

```
# Create dataset
customers <- data.frame(
  Customer_ID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)

# Display table
print(customers)
```

Customer_ID	Gender
1	Female
2	Male
3	Female

Set 8 Employee Performance Analysis

Dataset: Employee Performance

Employee ID Department Years of Service Performance Score

1 Sales 5 85

2 HR 3 92

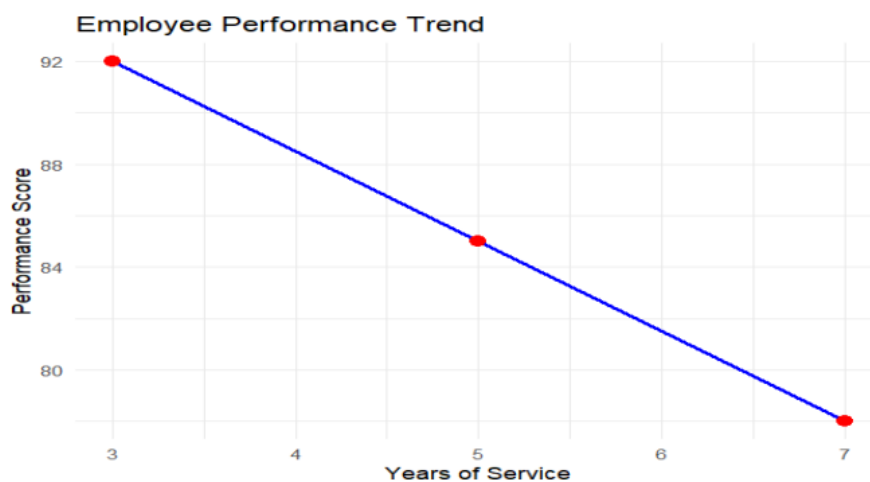
3 Marketing 7 78

1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

```
# Load library
library(ggplot2)

# Create dataset
employee <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)

# Line Chart
ggplot(employee, aes(x = Years_of_Service, y = Performance_Score, group = 1)) +
  geom_line(color = "blue", linewidth = 1) +
  geom_point(color = "red", size = 3) +
  labs(
    title = "Employee Performance Trend",
    x = "Years of Service",
    y = "Performance Score"
  ) +
  theme_minimal()
```

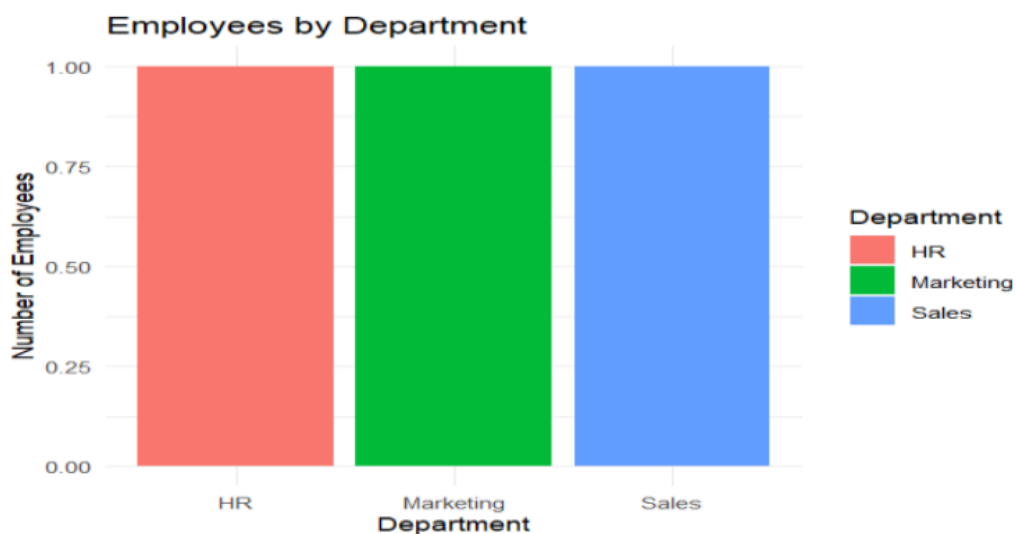


2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

```
# Load library
library(ggplot2)

# Create dataset
employee <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)

# Bar Chart
ggplot(employee,
  aes(x = Department,
    fill = Department)) +
  geom_bar() +
  labs(
    title = "Employees by Department",
    x = "Department",
    y = "Number of Employees"
  ) +
  theme_minimal()
```



3. Build a table to display the performance data for each employee. Label the table elements.

```
# Create dataset
employee <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)

# Display table
print(employee)
```

	Employee_ID	Department	Years_of_Service	Performance_Score
1	1	Sales	5	85
2	2	HR	3	92
3	3	Marketing	7	78

set 9

Dataset: Online_Learning_Activity.csv

Student_ID	Gender	Age	Course	Study_Time (hrs)	Videos_Watched	Quiz_Score	Login_Date
------------	--------	-----	--------	------------------	----------------	------------	------------

L01 Male 20 R 3.5 12 78 2025-01-05

L02 Female 22 R 4.2 15 85 2025-01-05

L03 Male 19 SQL 2.0 8 65 2025-02-08

L04 Female 21 R 5.0 18 92 2025-02-08

L05 Male 23 R 2.5 9 70 2025-03-12

L06 Female 20 SQL 4.0 14 88 2025-03-12

1. Import the dataset in R. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by

Course. Add title and labels. Interpret student performance.

```

# Load library
library(ggplot2)

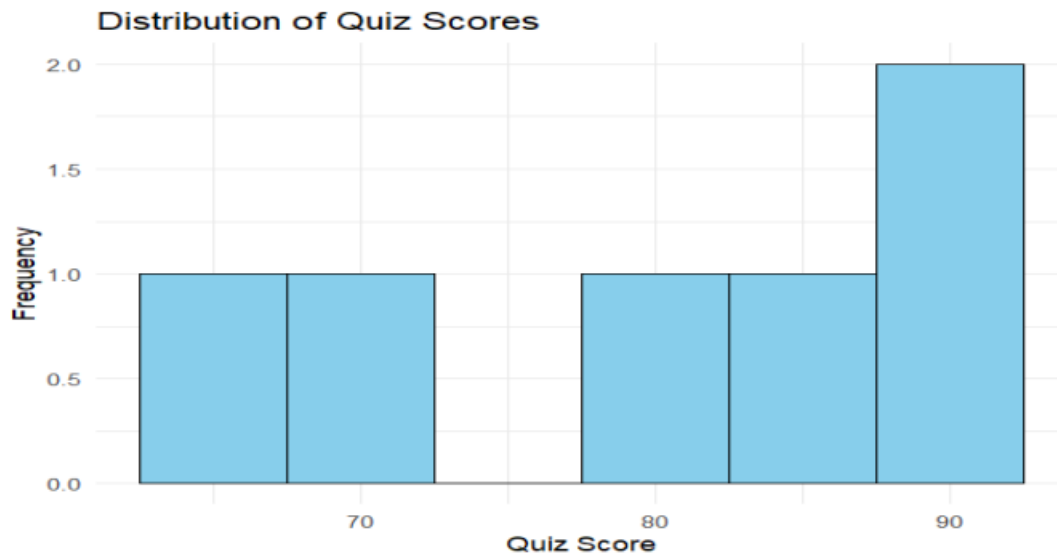
# Create dataset
learning <- data.frame(
  Student_ID = c("L01","L02","L03","L04","L05","L06"),
  Gender = c("Male","Female","Male","Female","Male","Female"),
  Age = c(20,22,19,21,23,20),
  Course = c("R","R","SQL","R","R","SQL"),
  Study_Time = c(3.5,4.2,2.0,5.0,2.5,4.0),
  Videos_Watched = c(12,15,8,18,9,14),
  Quiz_Score = c(78,85,65,92,70,88),
  Login_Date = c("2025-01-05","2025-01-05","2025-02-08",
"2025-02-08","2025-03-12","2025-03-12")
)

# Histogram
ggplot(learning, aes(x = Quiz_Score)) +
  geom_histogram(binwidth = 5, fill = "skyblue", color = "black") +
  labs(
    title = "Distribution of Quiz Scores",
    x = "Quiz Score",
    y = "Frequency"
  ) +
  theme_minimal()

# Boxplot
ggplot(learning,
  aes(x = Course,
    y = Quiz_Score,
    fill = Course)) +
  geom_boxplot() +
  labs(
    title = "Quiz Scores by Course",
    x = "Course",

```

```
y = "Quiz Score"
) +
theme_minimal()
```



2. Create a scatter plot using R programming between Study_Time and Quiz_Score. Represent Videos_Watched using bubble size. Apply transparency. Explain the learning pattern.

```
# Load library
library(ggplot2)

# Create dataset
learning <- data.frame(
  Student_ID = c("L01","L02","L03","L04","L05","L06"),
  Study_Time = c(3.5,4.2,2.0,5.0,2.5,4.0),
  Quiz_Score = c(78,85,65,92,70,88),
  Videos_Watched = c(12,15,8,18,9,14)
)

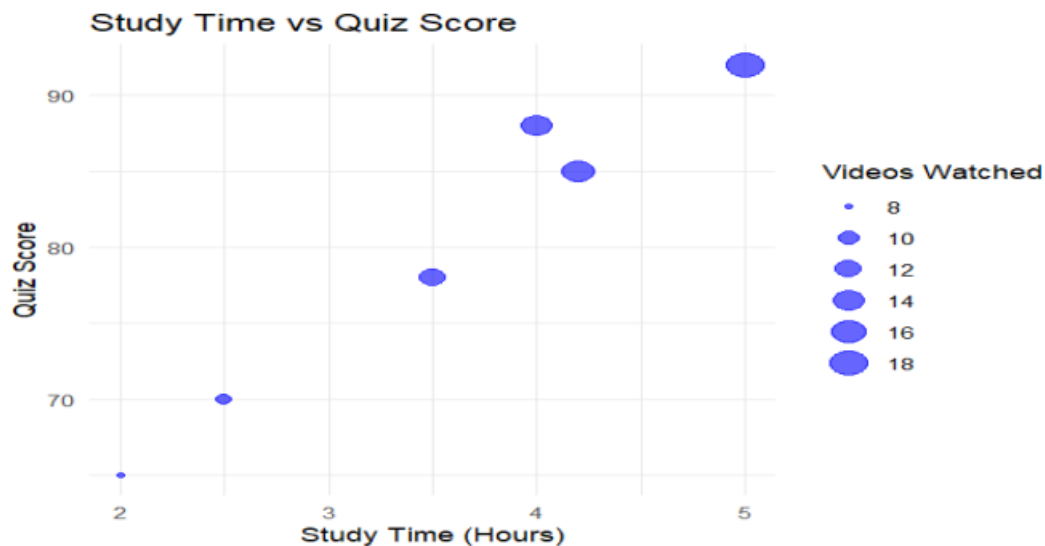
# Bubble Scatter Plot
ggplot(learning,
  aes(x = Study_Time,
    y = Quiz_Score,
    size = Videos_Watched)) +
  geom_point(alpha = 0.6, color = "blue") +
  labs(
```



```

title = "Study Time vs Quiz Score",
x = "Study Time (Hours)",
y = "Quiz Score",
size = "Videos Watched"
) +
theme_minimal()

```



3. Using R programming, convert Login_Date into Date format. Calculate average quiz score per month. Plot a line chart. Apply moving average smoothing. Identify the trend.

```

# Load libraries
library(ggplot2)
library(dplyr)

# Create dataset
learning <- data.frame(
  Student_ID = c("L01","L02","L03","L04","L05","L06"),
  Quiz_Score = c(78,85,65,92,70,88),
  Login_Date = c("2025-01-05","2025-01-05","2025-02-08",
    "2025-02-08","2025-03-12","2025-03-12")
)

# Convert to Date
learning$Login_Date <- as.Date(learning$Login_Date)

# Extract Month

```

```

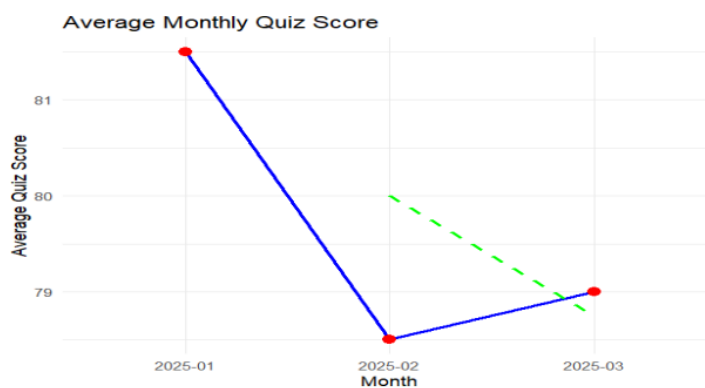
learning$Month <- format(learning$Login_Date, "%Y-%m")

# Average Quiz Score per Month
monthly_avg <- learning %>%
  group_by(Month) %>%
  summarise(Average_Score = mean(Quiz_Score))

# Moving Average
monthly_avg$Moving_Average <- stats::filter(
  monthly_avg$Average_Score,
  rep(1/2, 2),
  sides = 1
)

# Line Chart
ggplot(monthly_avg, aes(x = Month, y = Average_Score, group = 1)) +
  geom_line(color = "blue", linewidth = 1) +
  geom_point(size = 3, color = "red") +
  geom_line(aes(y = Moving_Average),
    color = "green",
    linewidth = 1,
    linetype = "dashed") +
  labs(
    title = "Average Monthly Quiz Score",
    x = "Month",
    y = "Average Quiz Score"
  ) +
  theme_minimal()

```



Set 10 Survey Responses Analysis

Dataset: Survey Responses

Survey ID Question 1 Question 2 Question 3

1 A B C

2 B A D

3 C A B

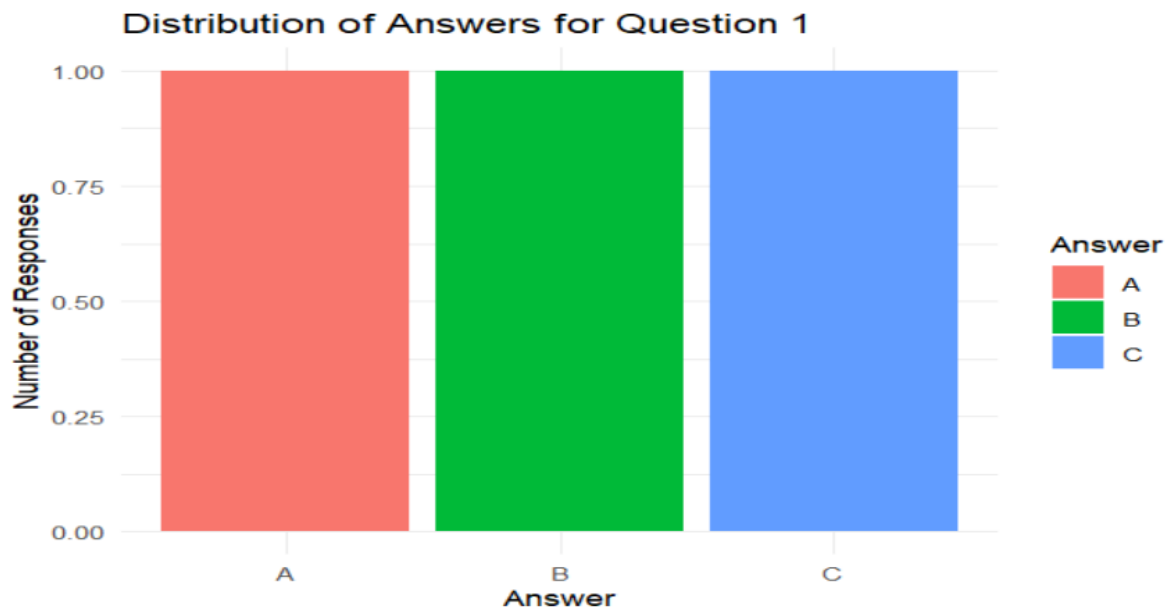
1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

```
# Load library
library(ggplot2)

# Create dataset
survey <- data.frame(
  Survey_ID = c(1, 2, 3),
  Question1 = c("A", "B", "C"),
  Question2 = c("B", "A", "A"),
  Question3 = c("C", "D", "B")
)

# Count responses for Question 1
q1 <- as.data.frame(table(survey$Question1))
colnames(q1) <- c("Answer", "Count")

# Grouped Bar Chart
ggplot(q1, aes(x = Answer, y = Count, fill = Answer)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Distribution of Answers for Question 1",
    x = "Answer",
    y = "Number of Responses"
  ) +
  theme_minimal()
```



2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

```
# Load libraries
library(ggplot2)
library(tidyr)

# Create dataset
survey <- data.frame(
  Survey_ID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)

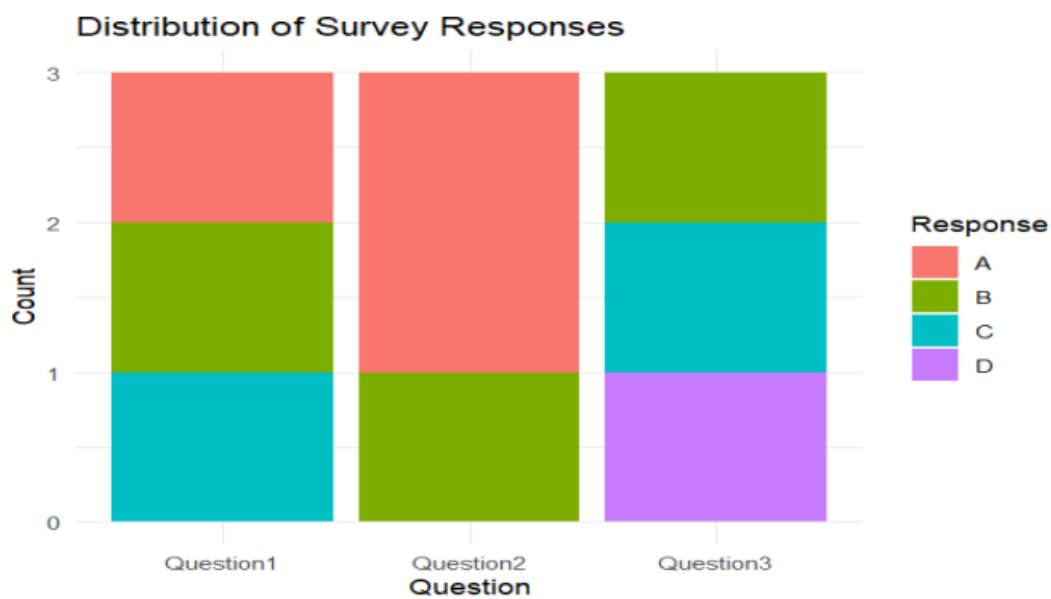
# Convert to long format
survey_long <- pivot_longer(
  survey,
  cols = c(Question1, Question2, Question3),
  names_to = "Question",
  values_to = "Response"
)

# Stacked Bar Chart
ggplot(survey_long,
```

```

aes(x = Question,
fill = Response)) +
geom_bar() +
labs(
title = "Distribution of Survey Responses",
x = "Question",
y = "Count"
) +
theme_minimal()

```



3. Build a table to show the survey response data for each survey. Label the table elements.

```

# Create dataset
survey <- data.frame(
  Survey_ID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)
# Display table
print(survey)

```

Survey_ID Question1 Question2 Question3

1	A	B	C
2	B	A	D
3	C	A	B

Set 11

Product Category Analysis

Dataset: Product Categories

Category Sales (in \$)

Electronics 50000

Clothing 35000

Appliances 40000

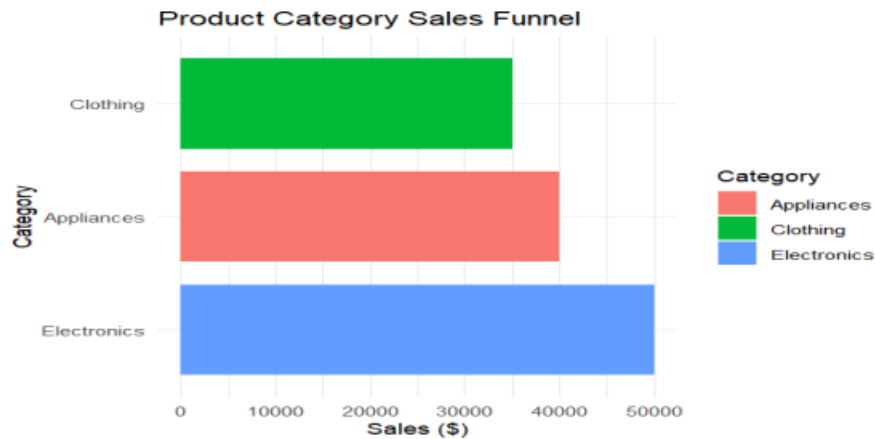
1.Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.

```
# Load library
library(ggplot2)

# Create dataset
category <- data.frame(
  Category = c("Electronics","Clothing","Appliances"),
  Sales = c(50000,35000,40000)
)

# Funnel Chart
ggplot(category,
  aes(x = reorder(Category, -Sales),
    y = Sales,
    fill = Category)) +
  geom_bar(stat = "identity", width = 0.8) +
  coord_flip() +
  labs(
    title = "Product Category Sales Funnel",
    x = "Category",
    y = "Sales ($)"
```

```
) +  
theme_minimal()
```



2. Build a table to display the sales data for each product category. Label the table elements.

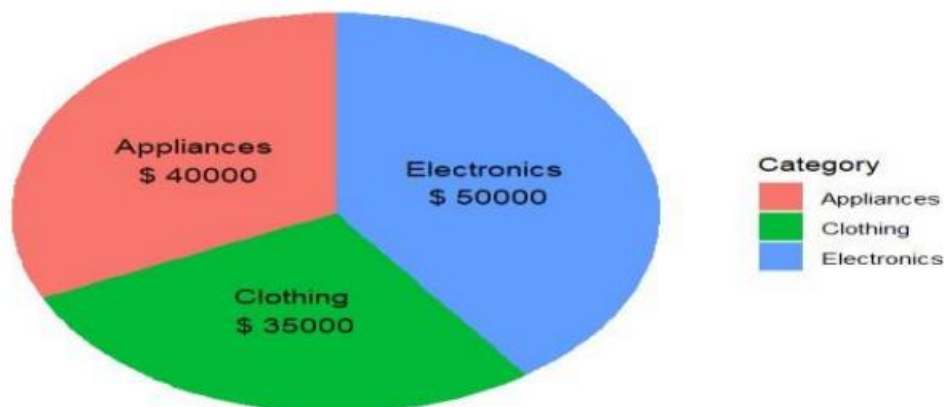
```
# Create dataset  
category <- data.frame(  
  Category = c("Electronics","Clothing","Appliances"),  
  Sales = c(50000,35000,40000)  
)  
# Display table  
print(category)  
Category Sales  
1 Electronics 50000  
2 Clothing 35000  
3 Appliances 40000
```

3. Create a pie chart to represent the distribution of sales across product categories. Include labels.

```
# Load library  
library(ggplot2)  
# Create dataset  
category <- data.frame(  
  Category = c("Electronics", "Clothing", "Appliances"),  
  Sales = c(50000, 35000, 40000)  
)
```

```
# Pie Chart
ggplot(category, aes(x = "", y = Sales, fill = Category)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar(theta = "y") +
  geom_text(
    aes(label = paste(Category, "\n$", Sales)),
    position = position_stack(vjust = 0.5),
    size = 4
  ) +
  labs(
    title = "Distribution of Sales Across Product Categories"
  ) +
  theme_void()
```

Distribution of Sales Across Product Categories



Set 12 :Dataset: Website Traffic

Date Page Views Click-through Rate

2023-01-01 1500 2.3%

2023-01-02 1600 2.7%

2023-01-03 1400 2.0%

1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

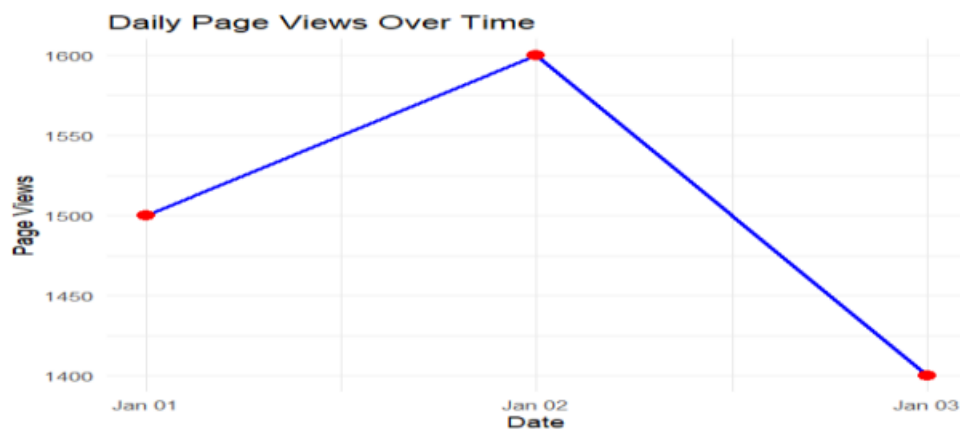

```

# Load library
library(ggplot2)

# Create dataset
traffic <- data.frame(
  Date = as.Date(c("2023-01-01",
    "2023-01-02",
    "2023-01-03")),
  Page_Views = c(1500,1600,1400),
  Click_Through_Rate = c(2.3,2.7,2.0)
)

# Line Chart
ggplot(traffic,
  aes(x = Date,
    y = Page_Views,
    group = 1)) +
  geom_line(color = "blue", linewidth = 1) +
  geom_point(color = "red", size = 3) +
  labs(
    title = "Daily Page Views Over Time",
    x = "Date",
    y = "Page Views"
  ) +
  theme_minimal()

```



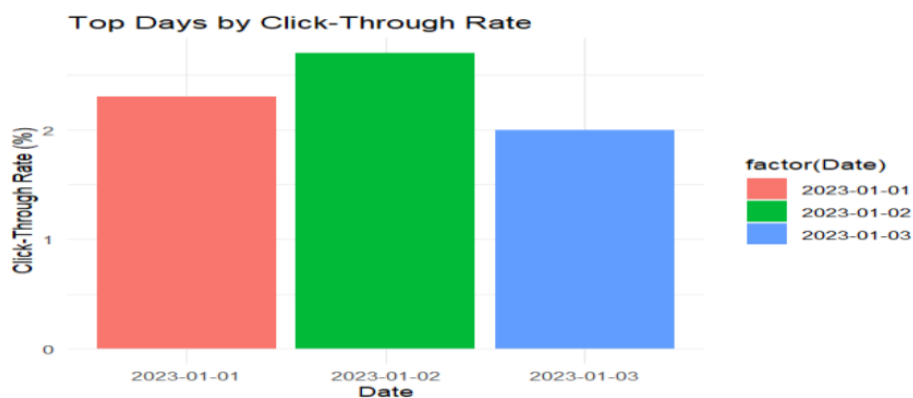
2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

```
# Load library
library(ggplot2)

# Create dataset
traffic <- data.frame(
  Date = as.Date(c("2023-01-01",
    "2023-01-02",
    "2023-01-03")),
  Click_Through_Rate = c(2.3,2.7,2.0)
)

# Sort data
top_days <- traffic[order(-traffic$Click_Through_Rate), ]

# Bar Chart
ggplot(top_days,
  aes(x = factor(Date),
    y = Click_Through_Rate,
    fill = factor(Date))) +
  geom_bar(stat = "identity") +
  labs(
    title = "Top Days by Click-Through Rate",
    x = "Date",
    y = "Click-Through Rate (%)"
  ) +
  theme_minimal()
```



3. Build a table to show the website traffic data for each day. Label the table elements.

```
# Create dataset
traffic <- data.frame(
  Date = c("2023-01-01",
           "2023-01-02",
           "2023-01-03"),
  Page_Views = c(1500,1600,1400),
  Click_Through_Rate = c("2.3%","2.7%","2.0%")
)
# Display table
print(traffic)
```

	Date	Click_Through_Rate
1	2023-01-01	2.3
2	2023-01-02	2.7
3	2023-01-03	2.0